Finite-Center Accountability and the Positive Route to Binary Rank

Live review note of June 20, 2026

Abstract

Two finite-center obstruction notes show that quotient-visible operational, sector, modular, recovery, continuum, or entropy data do not by themselves reconstruct hidden finite central rank or the predicate $q = 2$. This short note states the matching positive criterion. For a supplied finite center $\mathcal{Z}_F \simeq \ell^\infty(F)$, let $V_F = \mathcal{Z}_F/\mathbb{C}1$ be its contrast space. A declared family of probes, effects, outcome statistics, sector labels, recovery tasks, modular records, finite-alphabet laws, or direct measurements generates a readout map $R : V_F \to W$. The supplied data determine exactly the quotient $V_F/\ker R$. Hence full finite central cardinality is recoverable only when R is faithful on the physical finite-center contrast space and an admissibility or maximality premise says this represented center is the whole physical finite center. Under those premises $q = \operatorname{rank} R + 1$. A binary conclusion $q = 2$ additionally needs one accounted contrast plus a theorem or measurement identifying that contrast as the full finite center rather than a quotient of a larger hidden extension.

This is deliberately demotion-ready. In finite statistical language it is a Blackwell/sufficiency statement. In operator-algebraic language it is a Petz-style relative-to-the-supplied-channel boundary. The possible value is only the translation into black-hole/AQFT reconstruction: recovery, entropy, sector, modular, or locally covariant records force q or $q = 2$ only after finite-center accountability has been supplied.

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1 Question

Positive reconstruction theorems reconstruct the object supplied to them: a represented von Neumann algebra, a locally covariant functor, a sector category, a correctable code algebra, a channel tested by specified preparations and effects, or a Type II algebra with trace and normalization data. The finite-center obstruction is the converse question. If the displayed data have already passed through a quotient or coarse readout, what extra premise is needed before hidden finite central rank, and in particular the binary predicate $q = 2$, becomes reconstructible?

2 Finite Accountability

Definition 1 (finite-center contrast and readout)**.** Let F be a finite set with $q = |F|$, and let

$$\mathcal{Z}_F = \ell^\infty(F), \qquad V_F = \mathcal{Z}_F/\mathbb{C}1.$$

Thus $\dim V_F = q - 1$. A finite-center readout is a linear map

$$R : V_F \longrightarrow W$$

generated by the declared observation family: normal probes, effects, outcome statistics, sector records, recovery tasks, modular records, a finite-alphabet law, or a direct measurement channel. Its null directions are

$$N_R = \ker R.$$

Theorem 1 (finite accountability quotient)**.** *Let D_R be any data record whose dependence on finite-center contrasts factors through the readout map $R : V_F \to W$. Then:*

1. *D_R factors through V_F/N_R.*

2. *No invariant of the finite center that varies inside a coset of N_R is reconstructible from D_R.*

3. *The full finite central cardinality q is reconstructible from D_R only after adding both $N_R = 0$ and an admissibility or maximality premise saying that the represented finite center is the whole physical finite center, not a quotient, null sector, gauge reduction, excluded sector, or hidden extension.*

4. *Under those added premises,*

$$q = \dim(V_F/N_R) + 1 = \operatorname{rank} R + 1.$$

Proof. If two contrasts $v, v' \in V_F$ have the same image under R, then $v - v' \in N_R$. Every record built from R assigns the same value to v and v', so it factors through V_F/N_R. Consequently any target that changes within a coset of N_R is not a function of the supplied data.

The finite commutative algebra $\ell^\infty(F)$ has one atomic coordinate for each element of F. After constants are quotiented out, its contrast space has dimension $q-1$. If $N_R = 0$, the readout is faithful on that contrast space, so it determines $\dim V_F$. If the admissibility premise also says the represented finite center is physically exhaustive, this dimension is the physical finite-center contrast dimension, and $q = \dim V_F + 1$. If $N_R \neq 0$, or if the represented center is only a quotient or admissible subcenter of a larger hidden center, refinements inside the null or hidden directions change the unaccounted finite cardinality while preserving all R-visible data. $\square$

Corollary 1 (binary rank requires two inputs). *A conclusion $q = 2$ from finite-center readout data requires both:*

$$\dim(V_F/N_R) = 1$$

and a maximality, admissibility, direct-measurement, or finite-alphabet theorem identifying that accounted one-dimensional contrast as the full physical finite center. One accounted contrast alone proves only that the declared experiment sees a binary quotient.

3 Relation to Standard Sufficiency

The theorem should be demoted if a standard source already states this exact translation. Blackwell comparison of experiments makes sufficiency relative to a declared experiment or decision problem [1, 2]. Petz sufficiency makes recovery and equality of relative entropy relative to a declared subalgebra or channel [3, 4]. Von Neumann algebra ideal theory says ultraweakly closed two-sided ideals are central-projection summands [5, 6]. None of these facts is new.

The possible finite-center contribution is the reconstruction boundary: black-hole/AQFT records determine a finite central cardinality only when their hypotheses include faithful finite-center accountability. Without that premise, quotient-visible entropy, recovery, sector, modular, or locally covariant records reconstruct the supplied quotient, not hidden cardinality.

4 Claim Boundary for Reviewers

Claim	Status
R determines $V_F/\ker R$.	Elementary finite-dimensional linear algebra.
R determines full q when $\ker R = 0$ and the represented center is physically exhaustive.	Positive accountability criterion.
One accounted central contrast implies $q = 2$.	False without maximality, admissibility, direct measurement, or a finite-alphabet theorem.
Blackwell/Petz sufficiency may already contain the criterion.	Reviewer should identify the exact theorem if so; Team 1 should demote.

AQFT/OAQEC hypotheses may already supply maximality or faithful support.	Reviewer should name the hypothesis and decide whether it is present in the Team 1 statement.

5 Reviewer Question

Is Theorem 1 merely finite Blackwell/Petz sufficiency under a different name, or does the black-hole/AQFT translation isolate a useful claim boundary? If it is standard, the desired response is the citation and the exact theorem name. If it is false or too broad, the desired response is a counterexample or missing physical hypothesis.

References

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